Volumes by the Disc and Washer Methods

ANSWER KEY

Section 1: The Disc Method Around Coordinate Axes

1 Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{16\pi}{3}$

Disc formula: $V = \pi\int_0^4 [R(x)]^2 dx = \pi\int_0^4 (\sqrt{x})^2 dx = \pi\int_0^4 x dx = \pi[\frac{x^2}{2}]_0^4 = 8\pi$.

2 Find the volume when the region bounded by $y = x^2, x = 0,$ and $y = 4$ is revolved about the y-axis.

a. 8π ✓

b. 16π

c. 4π

d. $\frac{32\pi}{5}$

Revolving about y-axis: $x = \sqrt{y}$. $V = \pi\int_0^4 (\sqrt{y})^2 dy = \pi\int_0^4 y dy = \pi[\frac{y^2}{2}]_0^4 = 8\pi$.

3 What is the cross-sectional shape of a solid formed by the disc method?

a. **A solid circular disk of area** πR^2 ✓

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

The disc method generates perpendicular slices that are solid circular disks with area $A(x) = \pi[R(x)]^2$.

Section 2: The Washer Method Around Coordinate Axes

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4 The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$ ✓

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

Intersections at $x = 0, 1$. Outer radius $R(x) = x$, inner radius $r(x) = x^2$.
$V = \pi\int_0^1 (x^2 - x^4)dx = \pi[\frac{x^3}{3} - \frac{x^5}{5}]_0^1 = \pi(\frac{1}{3} - \frac{1}{5}) = \frac{2\pi}{15}$.

5 Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$ ✓

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$

Outer radius is $2x$, inner radius is x^2. Washer area is $\pi[(2x)^2 - (x^2)^2] = \pi(4x^2 - x^4)$.

6 Why is $\pi\int (R - r)^2 dx$ INCORRECT for the washer method?

a. **The area of a washer is $\pi R^2 - \pi r^2$, which is NOT equal to $\pi(R - r)^2$** ✓

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

Area of an annulus is $\pi(R^2 - r^2) = \pi R^2 - \pi r^2$. Squaring the difference $(R - r)^2 = R^2 - 2Rr + r^2$ introduces an erroneous $-2Rr$ term.

Section 3: Revolutions Around Non-Axis Horizontal Lines

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7 The region bounded by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. What is the outer radius $R(x)$?

a. $R(x) = 4 - x^2$ ✓

b. $R(x) = x^2$

c. $R(x) = 4$

d. $R(x) = x^2 - 4$

The axis of revolution is $y = 4$. The distance from the curve $y = x^2$ to the axis $y = 4$ is $4 - x^2$, which forms solid disks with radius $4 - x^2$.

8 Find the volume in the above problem revolved about $y = 4$.

a. $\frac{256\pi}{15}$ ✓

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

$V = \pi\int_{-2}^{2}(4 - x^2)^2 dx = 2\pi\int_{0}^{2}(16 - 8x^2 + x^4)dx = 2\pi[16x - \frac{8x^3}{3} + \frac{x^5}{5}]_0^2 = 2\pi[32 - \frac{64}{3} + \frac{32}{5}] = 2\pi[\frac{480 - 320 + 96}{15}] = 2\pi$

9 The region between $y = \sqrt{x}, y = 0,$ and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ **and** $r(x) = 1$ ✓

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$

Outer curve is $y = \sqrt{x}$, distance to $y = -1$ is $\sqrt{x} - (-1) = \sqrt{x} + 1$. Inner boundary is $y = 0$, distance to $y = -1$ is $0 - (-1) = 1$.

Section 4: Volumes with Known Cross-Sections

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10 The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$ ✓

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$

Side of square is $s(x) = 1 - x^2$. Area $A(x) = (1 - x^2)^2 = 1 - 2x^2 + x^4$.
$V = \int_{-1}^{1}(1 - 2x^2 + x^4)dx = 2[x - \frac{2x^3}{3} + \frac{x^5}{5}]_0^1 = 2(1 - \frac{2}{3} + \frac{1}{5}) = 2(\frac{8}{15}) = \frac{16}{15}$.

11 If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$ ✓

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

Radius is $r = \frac{s(x)}{2}$. Semicircle area is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(\frac{s}{2})^2 = \frac{\pi}{8}s^2$.

12 If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$ ✓

b. $A(x) = \frac{1}{2}[s(x)]^2$

c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

The area of an equilateral triangle with side s is $\frac{\sqrt{3}}{4}s^2$.

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